Adaptive, Variable-Order BDF for Coupled Reactor Transients

Accuracy and Performance on the LRA-2D Benchmark

GARNET benchmark report

19 August 2026

Abstract

This report evaluates the new adaptive, variable-order backward differentiation formula (BDF) implementation in `GARNET` on the LRA-2D rod-withdrawal benchmark. Accuracy is judged primarily against the original fine-mesh finite-difference reference data distributed with ANL-7416/2, not against a single modern code result. Cherezov, Vasiliev, and Ferroukhi's FEMCORE results are retained as a secondary independent comparison and as the algorithmic motivation for adaptive BDF. At 1 cm finite-volume resolution, the corrected `GARNET` model reproduces the static rods-in/rods-out eigenvalues to $-0.5/-14$ pcm. With cell-local heat and Doppler feedback, its six transient observables differ from the original reference by at most 1.90%; mean and peak assembly temperature differ by $+0.16\%$ and -0.06%. At comparable first-peak temporal error on the 3.75 cm mesh, automatic BDF5 is 5.10 times faster than adaptive backward Euler, uses 5.08 times fewer FGMRES applications, 4.80 times fewer inner iterations, and 15.8 times fewer accepted steps. At the paper's controls, `GARNET` takes 28.62 s on the documented development CPU; Cherezov et al. report 29.1 s for fourth-order FEMCORE but do not identify their CPU. The report therefore separates within-host speedups, raw cross-host timings, temporal convergence, and spatial/model convergence.

Principal findings

- The implementation performs genuine accept/reject control on a combined state of neutron flux, delayed precursors, and temperature, with rollback of every accepted-history component.
- Candidate $q-1$, q, and $q+1$ predictor defects are evaluated without additional diffusion solves. The order giving the largest safe next-step proposal is selected, with 5% hysteresis to limit chatter.
- At the paper's 10^{-5} tolerance, automatic order preserves the LRA history while reducing accepted steps from 461 to 414 and inner work by 3.2% relative to holding the maximum attainable order.
- Cell-local feedback follows the benchmark's space-dependent heat and Doppler equations, whereas Cherezov's element-wise temperature is an explicitly stated approximation. Refining the cell-local model to 1 cm puts all six transient observables within 1.90% of ANL.
- A matched-error comparison, rather than an equal-step or equal-tolerance comparison, gives a 5.10x wall-time and 4.80x inner-work advantage over backward Euler on the measured CPU.
- At comparable scalar spatial size and the same 10^{-5} tolerance, `GARNET` and fourth-order FEMCORE report 28.62 s and 29.1 s, respectively. This is a raw comparison only because the FEMCORE hardware is undisclosed.

1 Benchmark definition and reference hierarchy

LRA-2D is a two-group, two-delayed-family BWR quarter-core transient on an 11×11 lattice of 15 cm assemblies. The symmetry faces are reflective; the two exterior faces use zero flux. Axial leakage is represented by $B_z^2 = 10^{-4}$ cm^{-2}. Four assemblies in region R change linearly from region-3 to region-4 thermal absorption over $0 \le t \le 2$

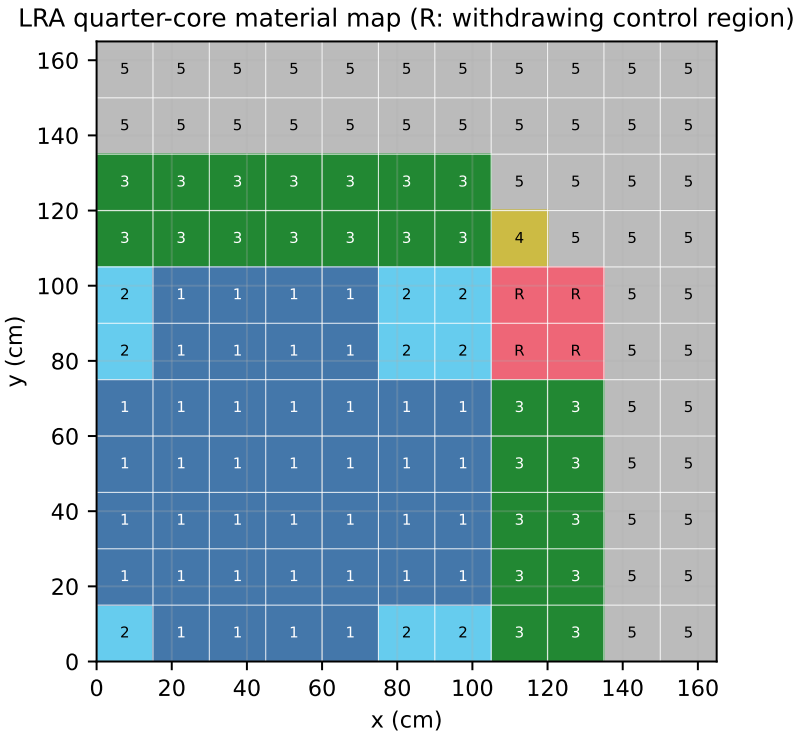


Figure 1: Quarter-core material map used by `GARNET`. Region R is the four-cell control region. Performance is measured with four FV cells per assembly edge (3.75 cm); the ANL convergence study extends to fifteen (1 cm).

s. Local adiabatic heating and a square-root temperature dependence of fast-group absorption provide the feedback. Initial average power is 10^{-6} W cm^{-3} and the initial temperature is 300 K.

The reference hierarchy matters. The "Reference" column in Cherezov et al.'s Table 6 is explicitly attributed to the fine-mesh finite-difference ANL-7416/2 dataset of Finnemann. Those values are the primary validation data in this report. FEMCORE is a secondary calculation using fourth-order B-spline finite elements and a related adaptive BDF algorithm. Agreement with FEMCORE alone would not establish agreement with the benchmark.

The source audit also resolved a significant input ambiguity. Cherezov's printed Table 4 lists both reflector absorptions one decade above the original ANL/DIF3D data. The corrected values are $\Sigma_{a1} = 6.034 \times 10^{-4}$ cm^{-1} and $\Sigma_{a2} = 1.911 \times 10^{-2}$ cm^{-1}. Using the printed thermal value while omitting axial buckling created an accidental rods-in cancellation and a false rods-out control-worth discrepancy. All calculations here use the original reflector data and the specified buckling. The Doppler multiplier is applied to physical fast absorption only, not to the $D_g B_z^2$ leakage term.

There are two feedback-resolution contracts in this report. Cherezov et al. state that they use one average temperature per finite element and note the inaccuracy introduced by that approximation. The `GARNET` "assembly" model reproduces this choice. The original benchmark equations instead define $T(t, \mathbf{x})$ and Doppler feedback pointwise. The `GARNET` "cell-local" model applies those equations on every FV cell and is the spatial-refinement path used for comparison to the fine-mesh ANL reference. Reported peak temperature is always the maximum *assembly-average* temperature, matching the Finnemann/Cherezov comparison field; the hotter maximum computational-cell temperature is retained in JSON as a separate diagnostic.

2 Adaptive BDF implementation

For order q, a nonuniform BDF step is written

$$\sum_{j=0}^{q} a_j u_{n+1-j} = h_n F(u_{n+1}), \tag{1}$$

where the coefficients are recomputed from the actual accepted time nodes. The same recurrence is used for each neutron group and delayed-precursor family. The coupled LRA temperature correction uses the identical selected order and time-node history.

The accepted-state vector for error control is

$$y = [\phi_1, \phi_2, C_1, C_2, T]\,. \tag{2}$$

Given a corrected endpoint y_c and a polynomial endpoint prediction y_p, the normalized defect is

$$e_h = \left(\frac{\sum_i \|y_{c,i} - y_{p,i}\|_2^2}{\sum_i \|\mathrm{atol} + \mathrm{rtol}|y_{c,i}|\|_2^2} \right)^{1/2}. \tag{3}$$

A step is accepted when $e_h \le 1$. Rejection restores flux, precursor, temperature, and every BDF history to the last accepted endpoint. The rejected attempt does not call the external accepted-step hook.

For an accepted order-q step, candidate predictor defects are formed for the available members of $\{q-1, q, q+1\}$. Each candidate proposes

$$h_{n+1}(r) = \frac{h_n}{1.25\,(e_r + 10^{-6})^{1/(r+1)}}, \tag{4}$$

bounded to a factor in $[0.2, 5]$. The candidate with the largest proposal is used unless its advantage over retaining q is less than 5%. Known discontinuities terminate a step exactly and restart all histories at order one. This estimator/order comparison reuses the converged endpoint and therefore adds vector operations but no spatial solve.

The paper-faithful rejection rule retries at $h_n/2$. An optional optimized rule applies the same defect exponent to a failed attempt, but caps the retry at $h_n/2$ and floors it at $0.2h_n$. Thus a marginal failure remains as conservative as the paper rule, while a very large defect can avoid a sequence of repeated halvings. Rejected endpoint, width, order, and work telemetry make this overhead directly auditable.

3 Experimental protocol

All runs use the corrected LRA definition, fully implicit endpoint constituent iteration, initial step 10^{-6} s, maximum step 0.1 s, and an exact event/restart at 2 s. The maximum BDF order is five. Performance and tolerance comparisons use the 3.75 cm mesh; the tight 10^{-6} BDF result is their fixed-model temporal reference. ANL spatial validation uses cell-local feedback at RTOL = 10^{-5} on 15, 7.5, 3.75, 1.875, and 1 cm meshes. A tight cell-local 3.75 cm run verifies that 10^{-5} contributes only 0.059% error to the first peak and negligible error to the remaining reported values.

Three distinctions are maintained throughout:

1. **Benchmark accuracy**: difference from the original ANL reference.
2. **Temporal error**: difference from the tight `GARNET` BDF solution on the same spatial model.
3. **Performance**: wall time and solver work at comparable temporal error, not at equal step width or equal controller tolerance.

Wall time is measured on one development CPU and is not portable. Accepted steps, FGMRES operator applications, and preconditioner inner iterations are reported as architecture-independent work counters. Each history and all telemetry used in the figures are stored as JSON beside this document.

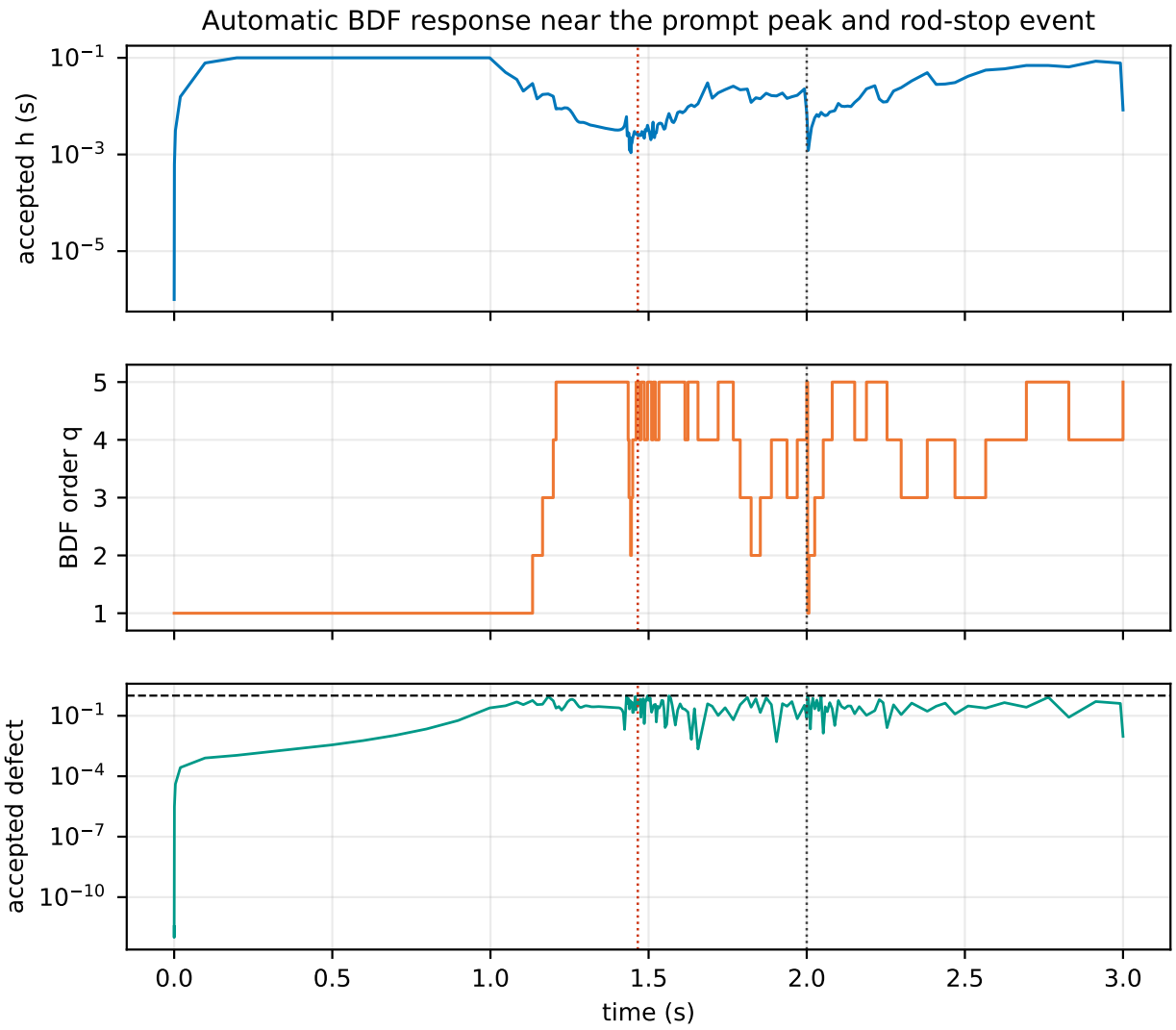


Figure 2: Accepted step width, selected order, and normalized accepted defect for the practical automatic-BDF run (RTOL = 10^{-3}). The red dotted line is the first power peak; the black dotted line is the 2 s rod-stop event. The controller contracts to millisecond steps near both rapid changes and recovers large steps in smooth intervals.

4 Accuracy against the original ANL reference

4.1 Static endpoints

Quantity	ANL	GARNET 3.75 cm	GARNET 1 cm	FEMCORE
k_{eff}, rods in	0.996330	0.9962305	0.9963254	0.996390
k_{eff}, rods out	1.015460	1.0148158	1.0153202	1.015500

At 1 cm, the rods-in and rods-out differences are −0.5 and −14 pcm. The rods-in value also reproduces the archived 1 cm DIF3D result 0.996325. The endpoint sequence therefore demonstrates ordinary finite-volume mesh convergence and rules out a missing control worth.

4.2 Transient observables

Quantity	ANL	Assembly T, 3.75 cm	Cell T, 3.75 cm	Cell T, 1 cm	FEMCORE
First-peak time (s)	1.436	1.46978	1.46975	1.44432	1.439
First-peak power (W/cm^3)	5411	5601.56	5417.33	5397.67	5641
Second-peak power (W/cm^3)	784	807.20	779.03	791.99	820
Power at 3 s (W/cm^3)	96.2	98.8319	95.8367	98.0318	98.7
Mean temperature at 3 s (K)	1087	1084.68	1059.80	1088.75	1122
Peak assembly T at 3 s (K)	2948	2953.16	2861.89	2946.28	3068

The tight assembly-temperature result isolates temporal error but retains Cherezov's element-wise feedback approximation. Resolving the benchmark's local feedback changes the 3.75 cm first/second/tail power differences to $+0.12/-0.63/-0.38\%$. Refining that model to 1 cm gives differences of $+0.58\%$ in first-peak time, -0.25% in first-peak power, $+1.02\%$ in second-peak power, and $+1.90\%$ in tail power. Mean and peak assembly temperature differ by only $+0.16\%$ and -0.06%. FEMCORE's largest Table 6 difference is 4.59% at the second peak.

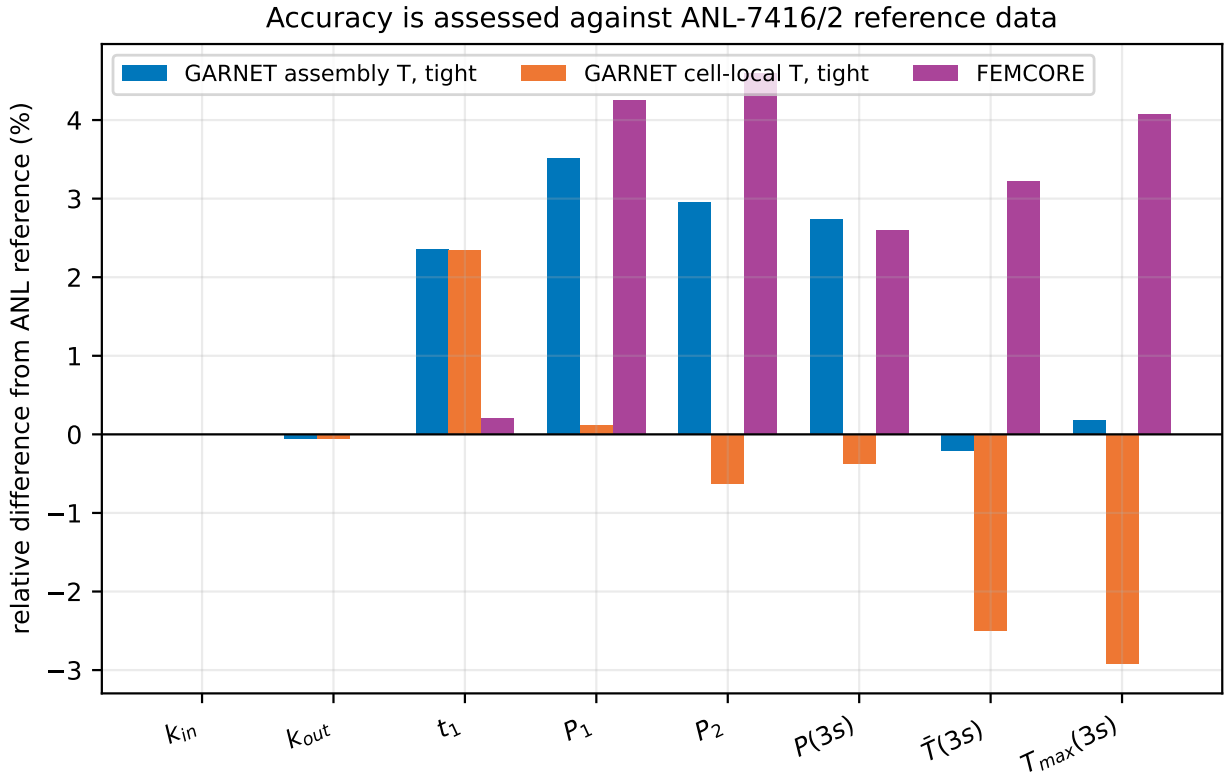


Figure 3: Feedback resolution at fixed 3.75 cm mesh. Cell-local feedback moves all three power values substantially closer to ANL; assembly-averaged feedback remains the appropriate reproduction of Cherezov's stated approximation. FEMCORE is secondary, not the reference baseline.

Figure 4 places the original point references on the computed histories. The practical assembly-temperature 10^{-3} BDF first peak nearly equals ANL by temporal/spatial error cancellation; the converged 1 cm cell-local history instead approaches the reference consistently across the full reporting set.

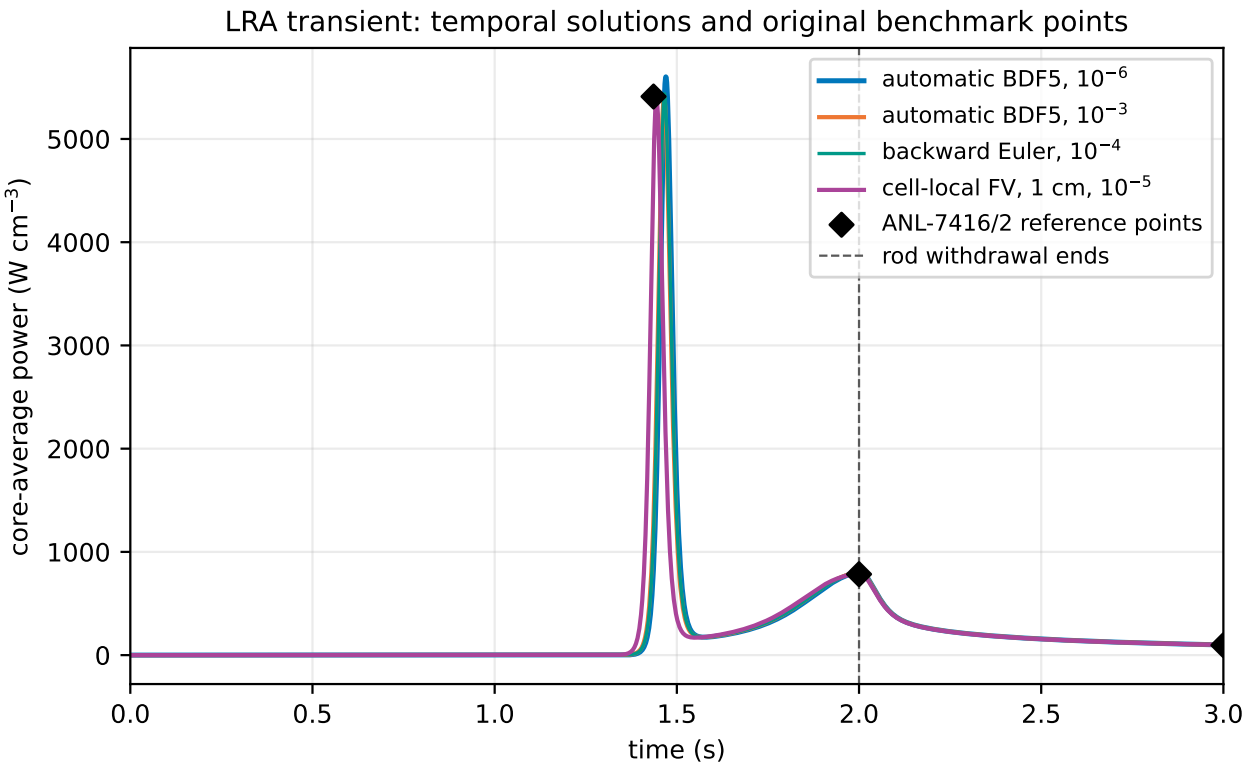


Figure 4: Complete accepted power histories with the three original ANL point references. The black line marks the end of linear control withdrawal. The 1 cm cell-local curve is the spatially refined validation result; the other curves support the fixed-mesh performance comparison.

4.3 Spatial convergence of the benchmark model

Cell (cm)	t_1	P_1	P_2	$P(3s)$	$\bar{T}(3s)$	$T_{max}(3s)$
15	−9.31 %	+10.38 %	+31.18 %	+17.25 %	+24.57 %	+50.54 %
7.5	+1.04 %	+1.65 %	+3.68 %	+1.39 %	+0.45 %	+3.56 %
3.75	+2.36 %	+0.06 %	−0.63 %	−0.38 %	−2.50 %	−2.92 %
1.875	+1.25 %	+0.20 %	+0.46 %	+1.10 %	−0.81 %	−1.22 %
1	+0.58 %	−0.25 %	+1.02 %	+1.90 %	+0.16 %	−0.06 %

The convergence is not monotone for every scalar—material interfaces, peak sampling, and feedback interact—but the envelope contracts decisively. At 1 cm all six transient values lie within 1.90% and both eigenvalues within 14 pcm. The hottest 1 cm computational cell reaches 3329 K; it is not compared to the 2948 K assembly-distribution reference and is stored separately.

5 Temporal convergence at fixed spatial model

RTOL	Acc./rej.	Inner its.	t_1 (s)	P_1	P_2	$P(3\,s)$
10^{-3}	199/22	301861	1.46589	5409.93	806.42	98.82496
10^{-4}	284/22	294134	1.46818	5563.33	807.08	98.82945
10^{-5}	414/29	341182	1.46997	5599.09	807.19	98.83172
10^{-6}	605/28	421475	1.46978	5601.56	807.20	98.83190

The first prompt peak controls the tolerance requirement. Relative to the 10^{-6} temporal result, its error is −3.42%, −0.68%, and −0.044% at 10^{-3}, 10^{-4}, and 10^{-5}. The second peak, tail power, and final temperatures converge

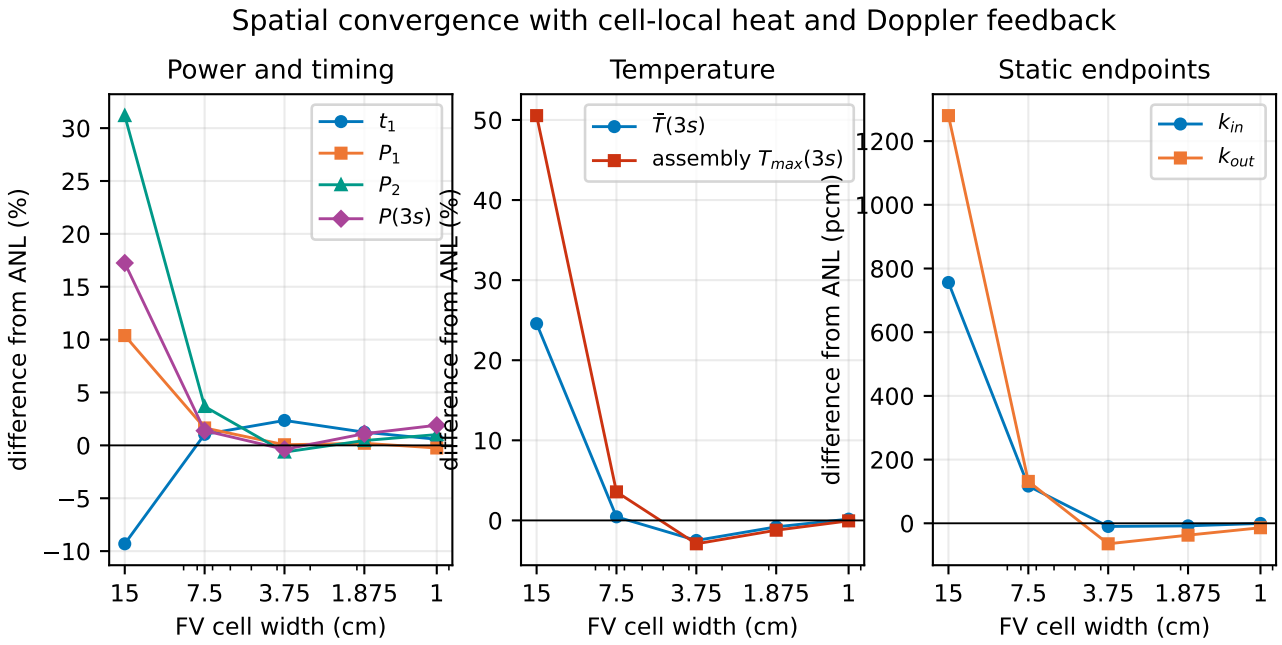


Figure 5: Signed spatial convergence against ANL using cell-local heat and Doppler feedback. Peak temperature denotes the maximum assembly average.

much sooner. The slightly smaller inner count at 10^{-4} than 10^{-3} is real: the looser run attempts more aggressive width/order changes and pays more rejected and constituent work.

The table uses the assembly-temperature track so that only temporal controls change. The cell-local check gives the same conclusion: tightening from 10^{-5} to 10^{-6} changes P_1 from 5414.13 to 5417.33 W/cm^3 (0.059%), while P_2, tail power, and both final temperatures change by less than 0.002%. Smaller steps converge the semi-discrete model; they cannot by themselves remove spatial or feedback-resolution error.

6 Performance of adaptive BDF

At the paper's 10^{-5} controls, automatic order gives 414 accepted and 29 rejected steps, compared with 461/21 when the maximum available order is held. The histories agree at reporting precision. Automatic selection cuts accepted steps by 10.2%, FGMRES applications by 3.4%, inner iterations by 3.2%, and measured time by 4.7%. The accepted count is close to the 399 regular steps reported for fourth-order FEMCORE; rejection counts are not directly equivalent because the state norms and spatial discretizations differ.

The main performance comparison uses the benchmark-faithful cell-local model and matches temporal accuracy. Relative to its tight BDF result, practical automatic BDF5 at 10^{-3} has -3.11% first-peak error. Backward Euler requires 10^{-4} to reach a comparable -2.92% error. Their first-peak-time errors are -0.35% and -0.27%; tail and temperature errors are smaller.

Method	RTOL	Acc./rej.	FGMRES	Inner	CPU (s)	P_1 error
Automatic BDF5	10^{-3}	201/30	13129	315381	27.54	$-3.11\,\%$
Backward Euler	10^{-4}	3180/19	66671	1512769	140.54	$-2.92\,\%$
BE/BDF ratio	—	15.82x	5.08x	4.80x	5.10x	comparable

This matched result is materially different from comparing methods at equal h or equal RTOL. Equal step widths strongly over-damp backward Euler near the prompt peak. Equal controller tolerances are also inappropriate because the predictor/corrector defect has a different order and error constant for each method.

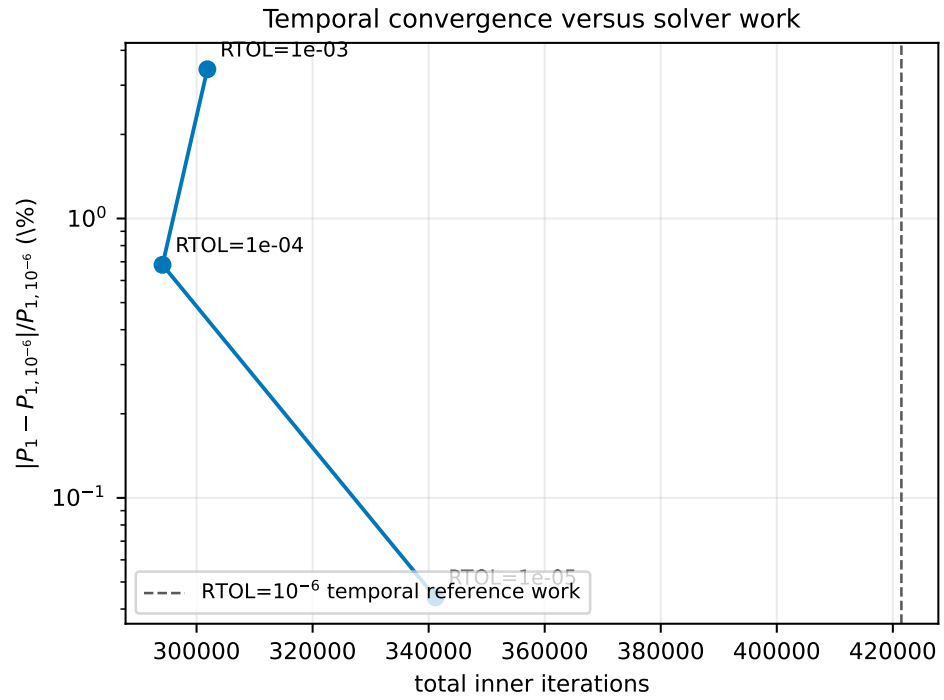


Figure 6: First-peak temporal convergence versus total inner iterations. The dashed line records the cost of the tight temporal reference; it is not plotted as a fictitious nonzero error point.

6.1 Rejected-step overhead

At the practical 10^{-3} tolerance, the optional error-scaled retry with a 0.5 cap retains 201 accepted steps but reduces rejected steps from 30 to 25. The following CPU values are medians of three sequential samples; deterministic work counters are identical across repetitions.

Rejection policy	Acc./rej.	FGMRES	Rej. FGMRES	Inner	CPU (s)
Factor two	201/30	13129	2523	315381	26.876
Error-scaled, cap 0.5	201/25	12395	2144	297831	25.494
Reduction	—	5.59%	15.0%	5.56%	5.14%

The optimized rule removes the two repeated-rejection streaks observed with factor-two control. Its first peak moves from 5248.71 to 5313.11 W cm^{-3}, closer to the ANL value of 5411 W cm^{-3}; tail and temperature observables change by at most 0.043%. At 10^{-5} the policies are work-neutral (14159 versus 14160 FGMRES applications), and first-peak time and power are identical at stored precision. A less conservative 0.8 cap was rejected: it induced a BDF1 reject/accept oscillation and increased the tight run to 623 accepted and 140 rejected steps. Error-scaled rejection therefore remains opt-in pending the HP-MR stability progression.

6.2 CPU comparison with Cherezov et al.

Cherezov et al.'s Tables 7–8 provide both a BDF work row and CPU time by FEM order. Their fourth-order accuracy calculation is the closest published spatial-size comparison to the 3.75 cm FV case: 2025 scalar degrees of freedom versus 1936 FV cells. Both use RTOL = 10^{-5} and automatic BDF up to order five.

Calculation	Spatial size	Acc./rej.	CPU (s)	Max. ANL diff.	Host
`GARNET` FV, 3.75 cm	1936 cells	410/27	28.62	2.92%	i5-1145G7
FEMCORE, FEM order 4	2025 DOF	399/7	29.1	4.59%	not reported
`GARNET` FV, 1 cm	27225 cells	391/29	614.50	1.90%	i5-1145G7

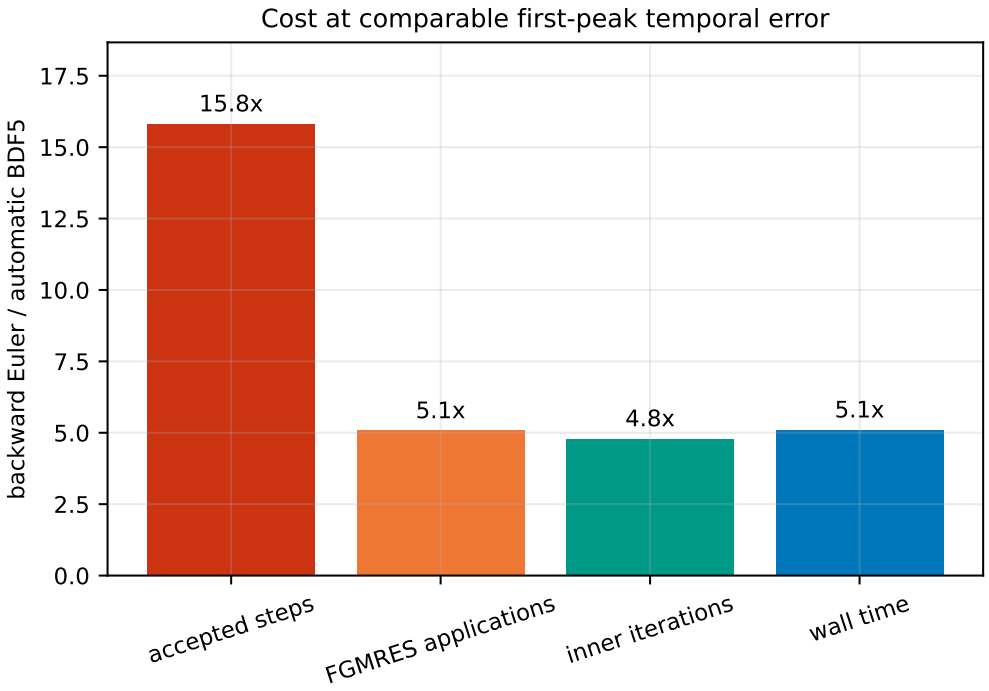


Figure 7: Backward-Euler cost divided by automatic-BDF5 cost at comparable first-peak temporal error. Work-counter ratios are more portable than the single-host wall-time ratio.

The 28.62 s and 29.1 s values are remarkably close, but they are *not* a cross-code speedup: FEMCORE's processor, thread configuration, compiler, and timing scope are not stated. The accepted-step comparison is more portable and also close. Solver iteration columns are not equated because FEMCORE counts block-GS/Anderson and ILU-CG operations, whereas `GARNET` counts monolithic FGMRES applications and preconditioner PCG iterations.

For completeness, the published FEMCORE CPU ladder for FEM orders 1–8 is 1.4, 4.7, 12.2, 29.1, 62.2, 121.7, 237.0, and 438.1 s. Figure 8 plots those raw times with the two `GARNET` measurements. It communicates computational scale, not normalized throughput.

6.3 Transferability of the LRA speedup

The fivefold LRA result is not a universal BDF-over-BE factor. A subsequent 11-group HP-MR slow-drum test at matched `RTOL`=10^{-3} gives 29/8 accepted/rejected steps for adaptive BE and 29/9 for automatic BDF5. BDF uses 23 order-one and six order-two steps; its 14.29 s coupled time is 5.3% above BE's 13.58 s, while its maximum power error is modestly smaller (0.0208% versus 0.0278%). The same conclusion holds in an accelerated-thermal feedback stress test.

The difference is temporal rather than spatial-solver throughput. LRA's sharp prompt and rod-stop peaks require 3180 BE steps for the accuracy reached by 201 BDF steps. The mild HP-MR maneuver is nearly linear between control knots, so the full-state selector correctly remains at orders one and two and adaptive BE chooses essentially the same widths. Comparisons against a deliberately fine 64-step BE reference still show a 1.5–2.0x cost reduction, but that is an adaptive-versus-reference result, not evidence that high-order BDF beats BE at matched error.

A subsequent 3-D GPU gate confirms the same mechanism on a Tesla T4. The 11-group refinement-2 core with ten axial layers has 145,200 active flux unknowns. Against a 32-step fixed-BE reference, adaptive BE takes 24 accepted and seven rejected steps, while automatic BDF5 takes 22 accepted and nine rejected steps. BDF remains mostly low order (15 BDF1 and seven BDF2 steps), but has the smaller maximum power error (0.0260% versus 0.0301%) and a fivefold smaller final normalized flux-shape error.

Three interleaved short solves resolve the GPU preconditioner choice. Plain tolerance-based group PCG has a 2.310 s median. Adding energy Anderson raises it to 2.620 s and is rejected; four reduction-free polynomial sweeps

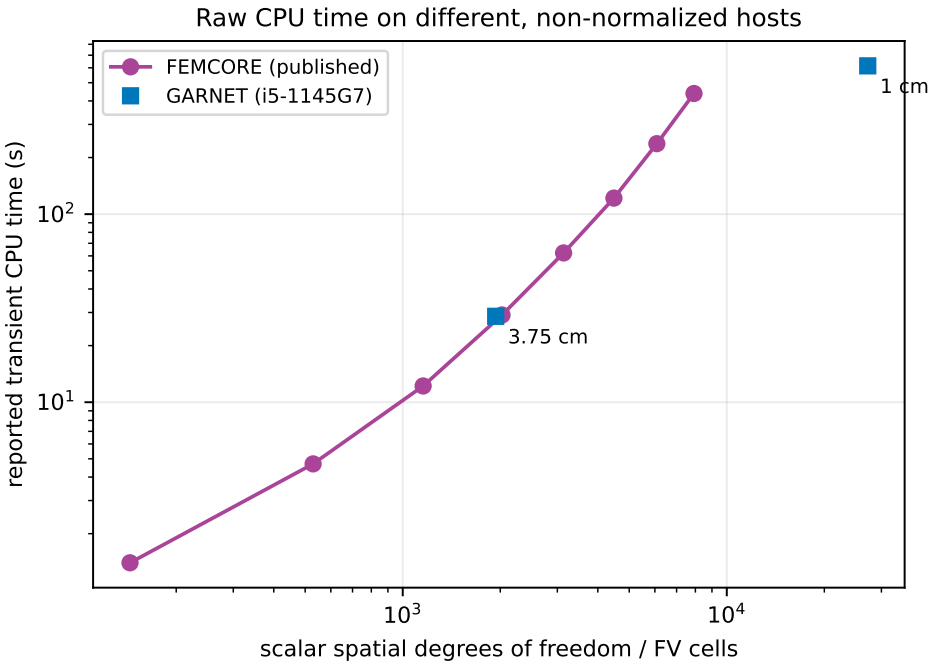


Figure 8: Published FEMCORE and measured `GARNET` transient CPU times. The hosts differ and FEMCORE's hardware is not disclosed; no cross-code speedup is formed.

with depth-one Anderson take 2.098 s, a 1.101x gain over plain PCG. Two complete runs with that polynomial configuration put the matched BDF/BE speed ratio at 0.946x and 0.902x: BDF is consistently 5.7–10.9% slower. In the latest run, BDF saves 88 outer applications on accepted attempts, but spends 286 extra on rejected attempts, for 2260 total versus 2062 for BE. The implementation gates pass: energy-group batching is active, persistent FGMRES storage is 61.27 MiB, and the latest one-transfer error norm and fused history predictor are 1.569x and 1.388x faster. The next acceleration target is therefore rejected-attempt Krylov work, not adaptive bookkeeping or another energy correction. Reference doubling and the production-size GPU case remain required.

7 Limitations and interpretation

- **Spatial method.** `GARNET` uses cell-centred finite volume; FEMCORE uses high-order B-spline FEM. The five-mesh ladder demonstrates the finite-volume envelope rather than assuming equivalence by nominal DOF.
- **Sparse reference history.** The original comparison contract provides peak and endpoint values rather than a digitized, uncertainty-rated continuous trace. The plotted black diamonds are those point references; lines between them are not invented.
- **Observable definition.** ANL/Cherezov spatial temperature values are assembly averaged. A cell-local simulation also has a hotter cell maximum; reporting it as the benchmark peak would mix observables.
- **Error cancellation.** Loose-tolerance and intermediate-mesh agreement can be fortuitous. Accuracy claims use a temporal check plus the full spatial sequence, not the closest isolated scalar.
- **Timing portability.** The within-host 5.08x FGMRES and 4.80x inner-work ratios are stronger evidence than seconds. Cherezov's CPU is unknown, so its raw time is not used to claim a speedup. The Tesla T4 quick gate is a same-device comparison. Repeated polynomial-preconditioned runs show BDF 5.7–10.9% slower than matched BE on the mild HP-MR trajectory; this is reported as a case-specific negative result, not a universal ratio.
- **Controller overhead.** Candidate-order defects require no spatial solve. The conservative error-scaled retry reduces practical-tolerance LRA rejection work, but the paper-control calculation still has more rejections than FEMCORE and uses a different state norm.

8 Conclusion

The corrected LRA model provides a demanding and useful validation case for adaptive time integration: it combines a prompt-supercritical peak, delayed neutrons, a moving control perturbation, a feedback-driven decline, a second peak at the rod-stop event, and strongly nonuniform useful time scales. The new `GARNET` BDF path resolves that history without instability, respects the full coupled-state error test and event rollback contract, and converges toward the original ANL data as the cell-local model is refined. At 1 cm all six transient checks are within 1.90%, temperature checks within 0.17%, and static endpoints within 14 pcm.

Performance is not inferred from equal time steps. At matched first-peak temporal accuracy, automatic BDF5 reduces portable linear-solver work by about a factor of five and measured CPU time by a factor of 5.10 relative to adaptive backward Euler. At paper controls and comparable spatial size, the raw `GARNET` and FEMCORE times are 28.62 and 29.1 s; the unknown FEMCORE hardware precludes a cross-code speedup claim. The combined accuracy and within-host performance evidence supports carrying the controller into the staged HP-MR transient cases. Error-scaled rejection provides a further 5.6% deterministic work reduction at practical tolerance, while factor-two rejection remains the paper-faithful default. The 3-D T4 quick gate preserves accuracy and validates the optimized GPU path, including a 1.101x repeated-median gain from its reduction-free polynomial preconditioner. It nevertheless shows no matched BDF-over-BE speedup on mild HP-MR motion because rejected BDF attempts dominate the saved accepted-step work. Adaptive BDF stays opt-in until reference convergence and production-scale HP-MR gates are complete.

Reproducibility

From the repository root, the structured aggregate benchmark is

```
PYTHONPATH=. python examples/lra2d_bdf_tolerance_benchmark.py --refine 4
```

The fine-mesh accuracy endpoint is reproduced with

```
PYTHONPATH=. python examples/lra2d_bdf_benchmark.py --refine 15 \
  --dt 1e-6 --order 5 --implicit-feedback --thermal-zones cell \
  --adaptive-rtol 1e-5 --automatic-order --include-static-endpoints
```

The complete-history JSON files were produced with `examples/lra2d_bdf_benchmark.py --include-history --output ...`. Run `make` in this report directory to regenerate figures and PDF from the stored data. The complete repository regression at this checkpoint is 575 passed, 5 skipped, and 13 deselected; the focused LRA/BDF subset is 82 passed.

References

[1] H. Finnemann, *Argonne Code Center: Benchmark Problem Book, Volume 2*, Argonne National Laboratory, 1977. DOI: `https://doi.org/10.2172/12030251`.

[2] A. Cherezov, A. Vasiliev, and H. Ferroukhi, "Application of Backward Differential Formula and Anderson's method for multigroup diffusion transient equation," *Annals of Nuclear Energy*, 210, 110837, 2025. DOI: `https://doi.org/10.1016/j.anucene.2024.110837`.

[3] Core Physics LLC, "LRA benchmark: archived fine-mesh DIF3D input and results," `https://corephysics.com/benchmarks/dif3d_lra.txt`, accessed 18 August 2026.